Practical: Bank Smart Contract Using Solidity

1. Aim

To write and deploy a smart contract using Solidity in Remix IDE that allows the bank owner to deposit ETH, withdraw ETH, and check the account balance.

2. Theory

A smart contract is a self-executing program stored on a blockchain. It automatically performs predefined operations when specified conditions are satisfied, without requiring an intermediary. In this practical, a Bank Smart Contract is developed using Solidity and deployed through Remix IDE. It provides basic banking operations such as depositing Ether, withdrawing Ether, and checking the available balance.

3. Functions Used

• **Constructor:** Runs once during deployment and stores the deployer's address as the account owner.
• **Deposite():** A payable function that accepts ETH and adds the received amount to the contract's internal balance.
• **Withdraw():** Checks the caller and available balance, reduces the recorded balance, and transfers ETH to the owner.
• **showBalance():** Returns the recorded balance without modifying blockchain state.

4. Important Solidity Keywords

Keyword	Description
pragma	Specifies the Solidity compiler version.
contract	Defines a smart contract.
uint256	Stores an unsigned 256-bit integer.
address	Stores an Ethereum address.
payable	Allows a function or address to receive ETH.
msg.sender	Represents the address that called the function.
msg.value	Represents the amount of ETH sent with a transaction, in Wei.
require()	Checks a condition and reverts if it is false.
constructor()	Executes once when the contract is deployed.
view	Specifies that a function does not modify blockchain state.

5. Working Procedure

1. Open Remix IDE.
2. Create a Solidity file named **bank.sol**.
3. Write the Bank Smart Contract code.
4. Compile the contract using Solidity compiler version 0.8.x.
5. Select Remix VM or connect MetaMask to the Sepolia test network.
6. Deploy the contract and confirm the transaction if using MetaMask.
7. Enter an amount and call **Deposite** to deposit test ETH.
8. Enter a withdrawal amount in Wei and call **Withdraw**.
9. Call **showBalance()** to view the remaining contract balance.

6. Advantages

• Transparent and traceable transactions.
• Reduces the need for intermediaries.
• Automatically validates deposits and withdrawals.
• Restricts banking operations using owner authentication.

7. Result

Successfully implemented and deployed a Bank Smart Contract using Solidity in Remix IDE. The contract supports ETH deposits, withdrawals, and balance checking with owner authentication.

References

1. Solidity Official Documentation
2. Remix IDE Documentation